# BETHELMIND DIGITAL SOLUTIONS

CAC CERTIFIED ENTERPRISE • RC 7034297 • LAGOS COMMERCIAL SUITE

**OFFICIAL SOLAR BOQ LOAD CERTIFICATE**

BM-SOLAR-BOQ-512677

VALIDATED ENGINEERING / COMMERCIAL OUTPUT

## Solar & Renewable Energy BOQ Engineering Specification

Clean Energy & Inverter Engineering

**VERIFIED SPECIFICATION**

### 1. EXECUTIVE MATHEMATICAL & FINANCIAL BENCHMARKS

RECOMMENDED INVERTER CAPACITY
**5.0 kVA / 48V Pure Sine Wave Hybrid**

LITHIUM BATTERY STORAGE (LIFEPO4)
**10.24 kWh (2x 5.12kWh 100Ah Rackmount)**

SOLAR PV ARRAY REQUIRED
**3.6 kWp (6x 600W Tier-1 Monocrystalline)**

ESTIMATED TURNKEY INSTALLATION
**₦4,850,000 NGN**

DISCO BAND A GRID TARIFF HEDGE
**₦185,000 / month**

DIESEL GEN PAYBACK PERIOD
**14.2 Months (vs ₦1,250/L PMS/AGO)**

### 2. DETAILED COMMERCIAL BILL OF QUANTITIES / LINE ITEM SCHEDULE

| SYSTEM COMPONENT | ENGINEERING SPECIFICATION | BRAND & WARRANTY LEVEL | ESTIMATED UNIT COST |
|---|---|---|---|
| Hybrid Solar Inverter | 5kVA 48V Pure Sine Wave, Dual MPPT 120-450VDC | Growatt / Deye (5 Years Warranty) | ₦1,450,000 |
| Lithium LiFePO4 Battery | 51.2V 200Ah (10.24kWh), 6000 Cycles @ 80% DoD | Felicity / Pylontech (5 Years Warranty) | ₦2,100,000 |
| Monocrystalline PV Panels | 6x 600W N-Type Bifacial Tier-1 High-Efficiency | Jinko / Longi (25 Years Linear Yield) | ₦780,000 |
| Protection & Accessories | AC/DC SPD, 125A DC Breakers, 16mm Grounding | Schneider / Suntree (Statutory Surge Kit) | ₦280,000 |
| Certified Installation | Roof mount, trunking, load balancing & testing | COREN / Bethelmind Certified Engineers | ₦240,000 |

### 3. NIGERIAN ROADSIDE & TECHNICAL FRAUD PREVENTION DIRECTIVES

| AUDIT POINT | THREAT LEVEL | INSPECTION DIRECTIVE & FRAUD MITIGATION ACTION |
|---|---|---|
| **Battery Chemistry Fraud Warning** | HIGH THEFT RISK | Never accept rebranded lead-acid batteries disguised in lithium metal casings. Inspect CAN-bus BMS telemetry via RS485 communication. |
| **Inverter Overloading Spike Check** | EQUIPMENT FAILURE | Calculate starting inductive surge from 1.5HP inverter ACs and deep freezers. Ensure inverter handles 2x surge for at least 5 seconds. |
| **DC Cable Undersizing Danger** | FIRE HAZARD | Enforce minimum 25mm2 to 35mm2 pure copper DC battery cables. Aluminum-clad copper (CCA) cables melt under 100A continuous draw. |

**OFFICIAL STAMPED VERIFICATION & GUARANTEE**

Calculated via Nigerian Trade & Market Indices • RC 7034297

Certified Date: 25 October 2026 • Settlement: Moniepoint MFB 6805375225

**BETHELMIND DIGITAL SOLUTIONS**

Technical Strategy & Commercial Desk

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